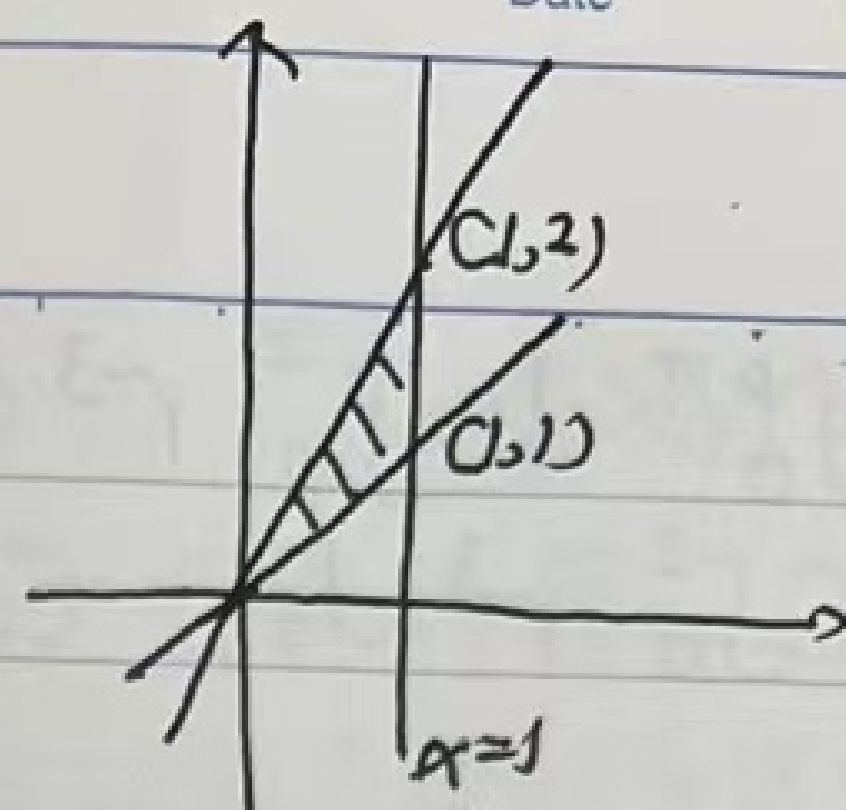


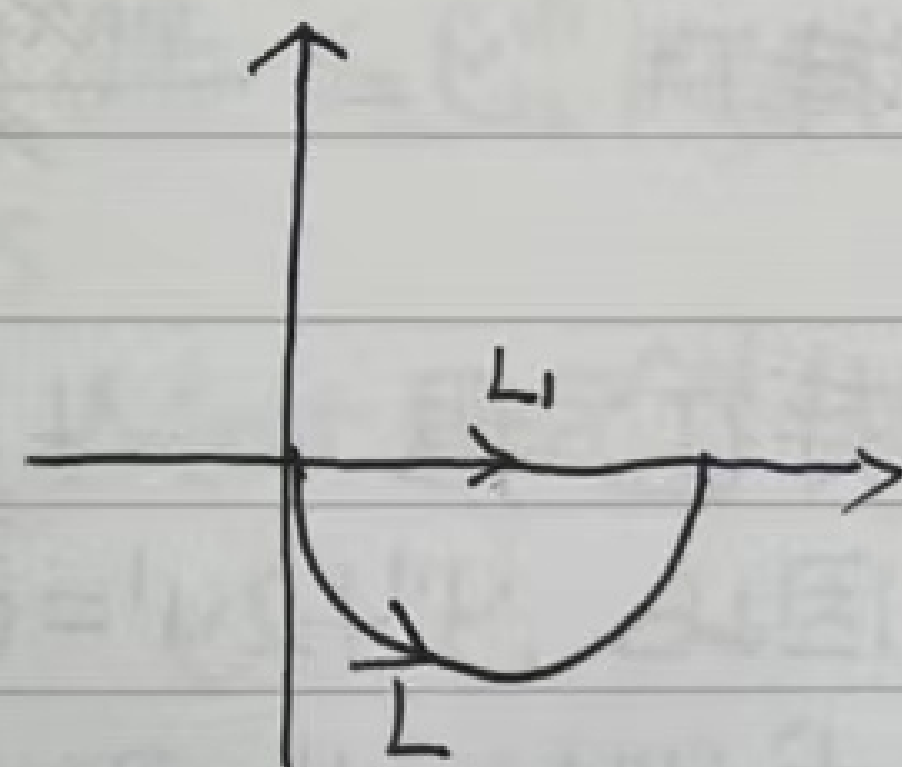
模拟题答案

$$\begin{aligned}
 \text{题1: } I &= \int_0^1 \left[\int_x^{2x} (x+y) dy \right] dx \\
 &= \int_0^1 \left(x \frac{y^2}{2} + \frac{y^2}{2} \Big|_{y=x}^{y=2x} \right) dx \\
 &= \int_0^1 \left(x \frac{2}{2} + \frac{3}{2} x^2 \right) dx \\
 &= \left(\frac{2}{5} x^{\frac{5}{2}} + \frac{1}{2} x^3 \right) \Big|_0^1 = \frac{9}{10}
 \end{aligned}$$



题2: 球坐标变换

$$\begin{aligned}
 I &= \iiint_{\Omega'} \frac{1}{e} e^2 \sin \varphi \, d\varphi \, d\theta \, d\phi \\
 \Omega' &= \{ (e, \varphi, \theta) : e \leq 2 \sin \varphi \cos \theta, \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \} \\
 I &= \iiint_{\Omega'} e \sin \varphi \, d\varphi \, d\theta \, d\phi \\
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{\pi} d\varphi \cdot \int_0^{2 \sin \varphi \cos \theta} e \sin \varphi \, de \\
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{\pi} 2 \sin^3 \varphi \cos^2 \theta \, d\varphi \\
 &= \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2 \cos^2 \theta \, d\theta \right) \cdot \left(\int_0^{\pi} \sin^3 \varphi \, d\varphi \right) \\
 &= 4 \cdot \frac{\pi}{4} \cdot 2 \cdot \frac{2}{3} = \frac{4\pi}{3}
 \end{aligned}$$



题3: 在 $x^2 + y^2 - 2x \leq 0$ 下半圆用路径无关性, 注意到

$$\partial \left(\frac{-1}{x-y+1} \right) = 2x \left(\frac{1}{x-y+1} \right)$$

因此可改为直径 L_1 上积分

$$I = \int_{L_1} \frac{dy - dx}{x - y + 1} = \int_0^2 \frac{dx}{x+1} = \ln 3$$

题4: 圆面 $S_1 = \{ z=1, x^2 + y^2 = 4 \}$

用 $S \cup S_1$ 围成区域 Ω'

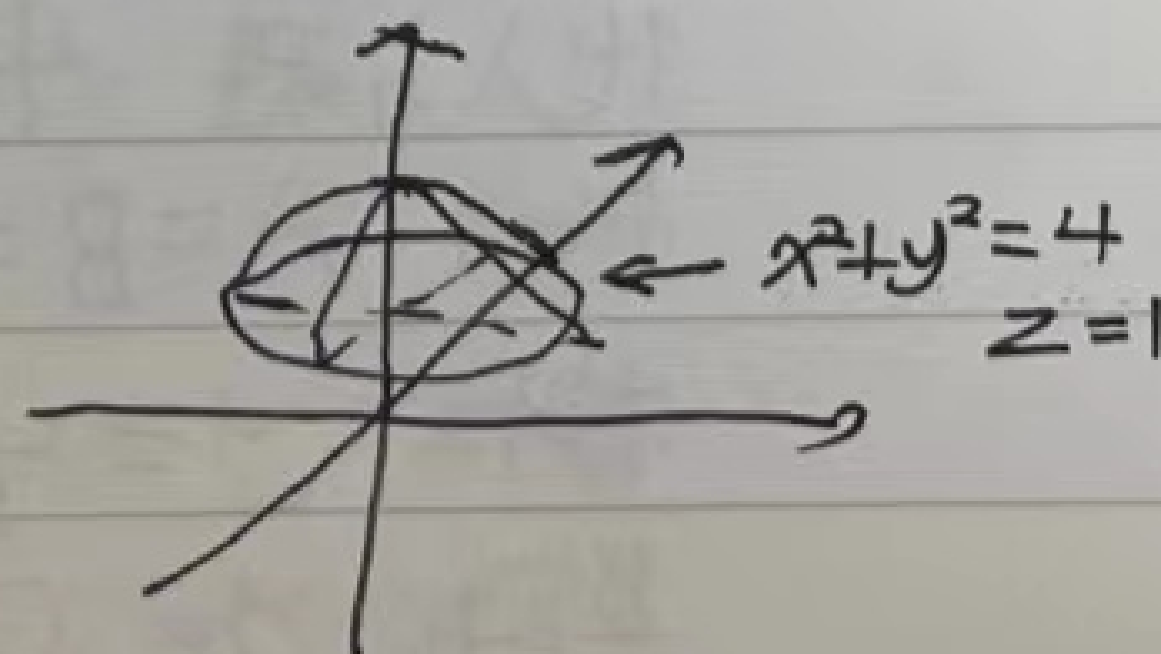
用 Gauss 公式

$$\iint_{S_{\text{外}}} + \iint_{S_1, \text{下}} = \iiint_{\Omega'} (y+x) \, d\sigma \quad \underline{\text{对称性}} \quad 0$$

$$I = - \iint_{S_1, \text{下}} y(x-z) \, dy \, dz + x^2 \, dz \, dx + (y^2 + x^2) \, dx \, dy$$

$$= \iint_{x^2+y^2 \leq 4} (y^2 + x^2) \, dx \, dy \quad (=\text{重积分})$$

$$= \iint_{x^2+y^2 \leq 4} y^2 \, dx \, dy \quad (x \text{ 部分有对称性})$$



$$= \int_0^{2\pi} d\theta \int_0^2 r^3 \cos\theta dr$$

$$= \left(\int_0^2 r^3 dr \right) \left(\int_0^{2\pi} \cos\theta d\theta \right) = 4\pi$$

题5: 一阶ODE $y' + \frac{2}{x} \cdot y = \frac{\sin x}{x}$

常数变易法, 对应齐次方程 $y' + \frac{2}{x} y = 0$

有通解 $y = C_0 e^{-2\ln x} = C_0 x^{-2}$

变易解 $y = C_0(x) \frac{1}{x^2}$

代入一阶ODE $C_0'(x) \cdot \frac{1}{x^2} = \frac{\sin x}{x}$

解得 $C_0'(x) = x \sin x$

即 $C_0(x) = -x \cos x + \sin x + C$

由此通解为 $y = \frac{1}{x^2} (C \sin x - x \cos x + C)$

代入 $y(\pi) = \frac{1}{\pi}$ 得 $\frac{1}{\pi} = \frac{1}{\pi^2} \cdot (\pi + C) \Rightarrow C = 0$

特解 $y = \frac{\sin x - x \cos x}{x^2}$

题6: 特征方程 $\lambda^2 + 2\lambda = 0 \Rightarrow \lambda = 0, -2$

因此 $y'' + 2y' = 0$ 有特解 $y = 1, y = e^{-2x}$

求 $y'' + 2y' = 3 + 4\sin 2x$ 特解. 设为 $y = \frac{3}{2}x + A \sin 2x + B \cos 2x$

$$y'' = -4A \sin 2x - 4B \cos 2x$$

$$y' = \frac{3}{2} + 2A \cos 2x - 2B \sin 2x$$

$$\text{代入方程 } 4\sin 2x = (-4A - 4B) \sin 2x + (4A - 4B) \cos 2x$$

$$\text{解得 } A = B = -\frac{1}{2}$$

$$\text{特解 } y = \frac{3}{2}x - \frac{\sin 2x + \cos 2x}{2}$$

$$\text{通解 } y = C_1 + C_2 e^{-2x} + \frac{3}{2}x - \frac{\sin 2x + \cos 2x}{2}$$

题7: 设第一卦限部分记作 Ω_1 , 体积为 V_1 , Ω_1 在 XOY 平面的投影为正方形 $[0, a] \times [0, a]$, 我们只需计算以 $[0, 0]$, $[a, 0]$, $[a, a]$ 为顶点的等腰直角三角形为底之部分, C 记为 D_1

$$V = 8V_1$$

$$= 16 \iint_{D_1} \sqrt{a^2 - x^2} dx dy$$

$$= 16 \int_0^a \left(\int_0^x \sqrt{a^2 - x^2} dy \right) dx$$

$$= 16 \int_0^a x \sqrt{a^2 - x^2} dx$$

$$= 16 \cdot \left(-\frac{1}{3} (a^2 - x^2)^{\frac{3}{2}} \right) \Big|_{x=0}^{x=a} = \frac{16}{3} a^3$$

题8: 用 Newton-Leibniz 公式

$$u_x = \frac{\partial u}{\partial x} \quad u_y = \frac{\partial u}{\partial y}$$

$$u(x, y) = \int_a^x u_x(t, y) dt$$

$$= \int_c^y u_y(x, s) ds$$

$$\text{因此 } |u(x, y)| \leq \int_a^x |u_x(t, y)| dt \leq \int_a^b |u_x(t, y)| dt$$

$$\text{且 } |u(x, y)| \leq \int_c^d |u_y(x, s)| ds$$

$$\text{二式相乘 } |u(x, y)|^2 \leq \left(\int_a^b |u_x(t, y)| dt \right) \left(\int_c^d |u_y(x, s)| ds \right)$$

对 $(x, y) \in D$ 积分 = 重积分

$$\iint_D |u(x, y)|^2 d\sigma \leq \left(\int_c^d dy \cdot \int_a^b |u_x(t, y)| dt \right) \cdot$$

$$\left(\int_a^b dx \cdot \int_c^d |u_y(x, s)| ds \right)$$

* 本题类型为 Sobolev 嵌入定理

$$\leq \left(\iint_D |u_x(t, y)| dt dy \right) \left(\iint_D |u_y(x, s)| dx ds \right)$$

题9: ① $\lambda = 0$ 时 $u \equiv 1$ 为本征函数

$$\text{② } a_1 = -1 \quad b_1 = 1 \quad a_2 = b_2 = 0$$

$$\begin{cases} u'' + \lambda u = 0 \\ u'(0) = u'(L) = 0 \end{cases}$$

解 = 边界值问题 $u'' + \lambda u = 0$ (对某固定 λ)

$$\text{ci) } \lambda = 0 \quad u = C_1 + C_2 x$$

$$\text{cii) } \lambda > 0 \quad u = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x$$

$$\text{ciii) } \lambda < 0 \quad u = C_1 e^{\sqrt{-\lambda} x} + C_2 e^{-\sqrt{-\lambda} x}$$

代入两个边界条件 $C_i) C_1 = \text{any } C_2 = 0 \Rightarrow \lambda = 0$ 为本征值 $u_\lambda = C$

$$C_{ii}) \begin{cases} \sqrt{\lambda} C_1 = 0 \\ \sqrt{\lambda} C_1 \cos \sqrt{\lambda} L - \sqrt{\lambda} C_2 \sin \sqrt{\lambda} L = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ C_2 \text{ 若非 } 0 \text{ 必 } \sqrt{\lambda} L = k\pi \end{cases} \quad k \in \mathbb{N}^*$$

本征值 $\lambda = (C \frac{k\pi}{L})^2$ 本征函数 $u_\lambda = C_2 \cos(C \frac{k\pi x}{L})$

$$C_{iii}) \begin{cases} \sqrt{\lambda} C_1 - \sqrt{\lambda} C_2 = 0 \\ \sqrt{\lambda} C_1 e^{\sqrt{\lambda} L} - \sqrt{\lambda} C_2 e^{-\sqrt{\lambda} L} = 0 \end{cases} \Rightarrow \begin{cases} C_1 = C_2 = 0 \\ \Rightarrow \lambda < 0 \text{ 无本征值} \end{cases}$$

综上: $\lambda = (C \frac{k\pi}{L})^2 \quad k \in \mathbb{N}^*$ 为本征值

对应本征函数 $u = C \cos(C \frac{k\pi x}{L})$

$a_1 = b_1 = 0 \quad a_2 = b_2 = 1$ 时间理分类讨论

得本征值 $\lambda = (C \frac{k\pi}{L})^2 \quad k \in \mathbb{N}$ 对应本征函数 $u = C \sin(C \frac{k\pi x}{L})$

③ 在方程左右同乘 $u(x)$ 在 $[0, L]$ 上积分

$$\int_0^L u''(x) u(x) dx + \lambda \int_0^L u^2(x) dx = 0$$

$$\text{而 } \int_0^L u''(x) u(x) dx = u(x) u'(x) \Big|_{x=0}^{x=L} - \int_0^L (u'(x))^2 dx \\ = u(L) u'(L) - u(0) u'(0) - \int_0^L (u'(x))^2 dx$$

$$u(L) u'(L) - u(0) u'(0) = (C \frac{b_2}{b_1}) \cdot (C \frac{a_2}{a_1}) u(L)^2 - (C \frac{a_2}{a_1}) u(0)^2$$

$$\text{代入得 } \underbrace{-\frac{b_2}{b_1} u(L)^2 - \frac{a_2}{a_1} u(0)^2}_{\leq 0} - \int_0^L (u'(x))^2 dx + \underbrace{\lambda \int_0^L u^2(x) dx}_{> 0} = 0$$

上式说明 $\lambda \geq 0$

$$\text{如 } \lambda = 0 \Rightarrow u(L) = u(0) = \int_0^L (u'(x))^2 dx = 0$$

$$\Rightarrow u(L) = u(0) = 0 \quad u' \equiv 0 \Rightarrow u \equiv 0 \text{ 与 } u \text{ 是本征函数}$$

综上 $\lambda > 0$

$$\textcircled{4} \int u_\lambda''(x) + \lambda u_\lambda(x) = 0 \quad u_\mu''(x) + \mu u_\mu(x) = 0$$

$$\text{分别乘以 } u_\mu \text{ 和 } u_\lambda \text{ 做积分 } \begin{cases} \int_0^L u_\lambda'' u_\mu dx + \lambda \int_0^L u_\lambda u_\mu dx = 0 \\ \int_0^L u_\lambda u_\mu'' dx + \mu \int_0^L u_\lambda u_\mu dx = 0 \end{cases} \quad (*)$$

$$\text{而 } \int_0^L u_\lambda'' u_\mu dx = u_\lambda u_\mu \Big|_{x=0}^{x=L} - \int_0^L u_\lambda' u_\mu' dx$$

$$= (u_\lambda' u_\mu - u_\lambda u_\mu') \Big|_{x=0}^{x=L} + \int_0^L u_\lambda u_\mu'' dx$$

Campus 得 $\int_0^L u_\lambda'' u_\mu dx = \int_0^L u_\lambda u_\mu'' dx = 0$ 代入 (*) 证知 $\mu = \lambda$ 与方程组 (*) 中 $\lambda \neq \mu$ 矛盾